

**MACARIO: A Narrative-Driven RPG for Teaching the Life and
Historical Role of Macario Sakay in Araling Panlipunan**

**A Capstone Project Proposal
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**In Partial Fulfilment
of the Requirements for the Degree
Bachelor of Science in Information Technology**

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ENDORSEMENT FORM FOR PROPOSAL DEFENSE

TITLE OF RESEARCH:

MACARIO: A Narrative-Driven RPG for Teaching The Life and Historical Role of Macario Sakay in Araling Panlipunan

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APPROVAL SHEET

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TABLE OF CONTENTS

	Page
Title Page	I
Endorsement form for Proposal Defense	II
Approval Sheet	III
Table of Contents	IV
List of Figures	V-VI
List of Tables	VI
Introduction	1
Project Context	1-2
Purpose and Description	3-4
Objectives	4-5
Scope and Limitations	5-7
Review of Related Literature/Studies/Systems	8-26
Methodology	27
Technical Background	27-29
Calendar of Activities	29-32
Resources Software/Hardware	32-34
Requirements Analysis	34-38
Requirements Documentation	38-39
Functional Requirements	39-42
Non-Functional Requirements	43-44
Storyboard Overview	45-46
Design of Software, System, Product, and/or Processes	46
References	56-59
Appendices	60-62
Resource Persons	63
Personal Technical Vitae	64-65

LIST OF FIGURES

Page	Figure Name	Figure No.
17	Gameplay Interface and Street Scene of Pompeii: The Mystery of Lucius.	1.1
18	Login Interface and Main Menu of Heroes of History by InnoGames.	1.2
19	Episode Selection and Story Modules of Mission US by PBS.	1.3
20	Discovery Tour Time Periods Selection Interface of Assassin's Creed Discovery Tour.	1.4
21	Album and Settings Interface of Galà, a Filipino Indie Educational Game.	1.5
22	Player Information and Title Screen of Hiraya: A Bookworm's Odyssey.	1.6
23	Character Profile and Historical Inspiration of Joe and Andrew in the Game.	1.7
38	Summarize Using Use-Case Diagram	2.1
46	Agile Scrum Development	3.1
48	Design of Software, System, Product, and/or Processes	3.2
50	Data Flow Diagram	3.3
52	Entity Relationship Diagram	3.4
54	Prototype (UI) Gameplay	3.5
54	Login Screen	3.6
55	Assessment Screen	3.7
55	Progress Screen	3.8

LIST OF TABLES

Page	Table Name	Table No.
29-33	Gantt Chart	1.0

PROJECT CONTEXT

The study is conducted at Imus National High School, a public secondary institution in Cavite established in 1971. The school serves a large student population, with class sizes typically ranging from 40 to 50 students per section. In this environment, teachers face high instructional demands while fulfilling curriculum requirements.

In Araling Panlipunan classes, instruction remains largely traditional, relying on lectures, textbook discussions, note-taking, and memorization. While these methods provide structured content delivery, they are predominantly passive and offer limited opportunities for active student engagement. Consequently, a disconnect exists between established teaching practices and the increasingly interactive, digital preferences of today's learners.

According to subject teacher Kimberly Donadillo-Espiritu, the most persistent challenge is the widespread perception of history as uninteresting among students. This perception negatively affects participation, attention span, and information retention. Although teachers occasionally supplement lessons with images, memes, or simplified explanations, usually through a projector, these strategies only generate short-term engagement and fail to produce sustained interest or improved comprehension.

In discussing potential solutions, the teacher specifically suggested Macario Sakay as a suitable figure for deeper instructional focus. She noted that Sakay's life offers a continuous, linear narrative; From Katipunan member to guerrilla leader during the Philippine-American War to eventual capture, which is rarely taught in depth due to time constraints. Yet, his role is explicitly included in the Grade 8 Araling Panlipunan curriculum as part of the Philippine-American War resistance, and his historical contribution, sustaining organized Filipino resistance after the fall of the First Philippine Republic, is significant enough to warrant dedicated attention. In her assessment, Sakay's story could anchor a more engaging and coherent learning experience, yet no existing supplementary material explores it properly.

To validate and contextualize this suggestion, the researchers conducted an exploratory survey among Grade 8 students from Section Diligence. A sample of 44 respondents was drawn from a population of 48 students. The results, though not generalized across all sections, provide strong indicative support for the teacher's insight: 77.3% perceive history as less engaging than other subjects; 81.8% spend most of their free time playing digital games; 70.5% prefer interactive or visual learning; and 72.7% report difficulty recalling historical information. More critically, 81.4% could not identify Macario Sakay or describe his role prior to the survey, while 83.7% expressed greater interest in learning history through the perspective of a specific historical figure.

These findings confirm two interrelated issues. First, students are generally disengaged from history as currently taught. Second, and more specifically, the teacher's suggestion about Sakay is validated: there exists a measurable knowledge gap regarding his life and role, despite his inclusion in the Grade 8 Araling Panlipunan curriculum. His story of leading organized guerrilla resistance after the fall of the First Philippine Republic receives insufficient attention, leaving students unaware of who he was and why he matters historically.

This gap is the central concern the study aims to address. Macario Sakay offers a complete, chronologically structured narrative that is difficult to convey effectively through lectures and memorization alone. However, it is well-suited for translation into interactive gameplay, where students learn by experiencing events through Sakay's perspective rather than merely recalling dates and facts.

Accordingly, the study focuses on developing a narrative-driven educational supplementary game centered specifically on the life and historical role of Macario Sakay. The objective is not only to increase student engagement but to directly address a documented instructional gap by presenting Sakay's story in a way that traditional methods cannot achieve.

PURPOSE AND DESCRIPTION

The proposed system, MACARIO, is a narrative-driven educational Role-Playing Game (RPG) designed to serve as a supplementary instructional tool for Araling Panlipunan. The system aims to enhance the delivery of historical content by integrating interactive gameplay with structured narrative progression, enabling students to engage with the life and historical role of Macario Sakay in an experiential learning environment.

The system is developed as a 2D side-scrolling RPG with a linear progression composed of four narrative acts, each act representing a major phase of Sakay's life. The acts are sequenced chronologically to maintain historical continuity, covering Sakay's journey from Katipunan member to guerrilla leader during the Philippine-American War, and finally to his capture and execution.

Gameplay mechanics include movement and jumping for navigation, interaction for dialogue and event triggers, stealth-based navigation for avoiding detection, melee combat for close-range encounters, and a special attack mechanic for limited firearm-based sequences. These mechanics are applied depending on the contextual requirements of each act to ensure alignment between gameplay and historical events.

The game features a dynamic difficulty system where the complexity of missions, stealth detection windows, and combat challenges increase progressively from Act I to Act IV. This ensures students are gradually challenged as they advance through Sakay's story.

Specific game scenarios include evading Guardia Civil patrols in Manila while distributing revolutionary pamphlets as a young Katipunan member, surviving a reconnaissance mission into the mountains to establish a hidden base for the Tagalog Republic, leading a guerrilla raid against colonial forces, and navigating the tense negotiation leading to Sakay's surrender under a false promise of amnesty.

Before each pre-test, the system presents a historical trivia fact related to Macario Sakay or the Philippine-American War period. This activates prior knowledge and provides context. The system then manages pre-test and post-test evaluations per act alongside in-

game performance scoring based on objective completion, stealth effectiveness, and combat efficiency.

A teacher's dashboard allows instructors to view student progress, act upon completion of status, assessment results, and average performance scores. A class-level overview of completion rates and assessment of averages is also provided.

The system is deployed as a web-based application optimized for Android mobile devices (accessible via standard browsers). Access is restricted through authenticated login credentials managed by the teacher. The game functions as a supplementary instructional material, not a replacement for traditional teaching.

The beneficiaries of this study include Grade 8 students, who will gain access to an interactive learning tool for understanding Sakay's historical role; Araling Panlipunan teachers, who will benefit from a structured system for monitoring performance; school administrators, who may utilize the system as an instructional innovation; and future researchers, who may use this study as a reference for game-based learning in history education.

OBJECTIVES OF THE STUDY

The objective of the study is to develop and evaluate MACARIO, a narrative-driven educational Role-Playing Game (RPG) designed as a supplementary instructional tool for Araling Panlipunan that improves the delivery of Macario Sakay's life and historical role through interactive gameplay, structured historical narrative, and integrated assessment mechanisms.

Specific Objectives:

- To design and develop a 2D narrative-driven RPG module that presents the life and historical role of Macario Sakay through a four-act structured linear progression and interactive gameplay mechanics.
- To implement a system module integrating gameplay mechanics (including dynamic difficulty progression, health system, equipment system, and cosmetic

reward system), scoring algorithms, and a teacher monitoring interface for tracking and recording student performance.

- To develop and integrate an assessment module that evaluates student learning and engagement through pre-test and post-test evaluations per act (each preceded by a historical trivia fact), combined with in-game performance scoring and user feedback.

SCOPE AND LIMITATIONS

SCOPE

This study focuses on the development and evaluation of a narrative-driven educational Role-Playing Game (RPG) that presents the life and historical role of Macario Sakay within an interactive learning environment. The system is developed using the Unity game engine and deployed as a web-based application optimized for Android mobile devices (accessible via Chrome browser) and PC web browsers. It follows a linear progression structure composed of four narrative acts (Act I: The Awakening, Act II: The Long Shadow of War, Act III: The Republic in the Shadows, Act IV: The Bitter Harvest). Each act corresponds to a major phase of Sakay's life and is sequenced chronologically to maintain historical accuracy.

A.

Students

The system provides students with an interactive learning environment that includes movement, jumping, stealth navigation, melee combat, a special attack system, a health system (health pool depleted by enemy attacks or hazards, restored via collectible items), an equipment system (equippable items that alter combat effectiveness), and a cosmetic reward system (earned in-game currency to unlock period-correct outfits). These mechanics are applied across four narrative acts with dynamic difficulty progression (mission complexity and challenge increase progressively from Act I to Act IV).

The system also includes structured story progression through dialogue sequences and scripted events to support historical context and immersion. Before each pre-test, a

historical trivia fact related to Sakay or the period is presented to activate prior knowledge. Students are required to complete pre-test and post-test assessments per act. Their learning progress is evaluated through in-game performance metrics based on objective completion, efficiency, stealth performance, and survival. Students also have access to saved progress per session, allowing them to continue their learning across multiple logins.

Teachers

The system provides teachers with tools for instructional monitoring and evaluation. This includes managing student access to the system, monitoring act progression, and viewing assessment results per student. Teachers can also review in-game performance scores and overall progress summaries to support classroom-level evaluation of learning outcomes. A class-level overview of completion rates and assessment averages is also provided to assist in identifying learning gaps.

B.

Gameplay Module

This module handles the core interactive system, including player movement, stealth mechanics, combat interactions, event-triggered sequences, health and equipment systems, and the cosmetic reward system. It also manages in-game progression and mission-based objectives aligned with the four-act narrative of Macario Sakay's life.

Narrative and Progression Module

This module manages the structured storytelling of the system, ensuring chronological presentation of Macario Sakay's life through four linear acts. It integrates dialogue systems and cutscenes to support narrative delivery from the perspective of Macario Sakay.

Assessment and Evaluation Module

This module handles student assessment through pre-test and post-test evaluations per act. Each pre-test is preceded by a historical trivia fact. It also includes automated scoring based on in-game performance metrics such as objective completion, efficiency, stealth, and survival, along with basic user feedback collection to measure engagement.

Teacher and Analytics Module

This module provides teachers with access to student performance data, including assessment results, progress tracking, completion status, average scores, and act completion rates for classroom-level monitoring.

Authentication and Access Module

This module manages secure login and user authentication through teacher-managed accounts or approved login systems. It ensures that only authorized students can access the system during supervised school-based sessions or designated review use.

Limitations

- The teacher monitoring interface is limited in its analytical capability, as it only presents basic summaries of student progress, assessment results, and in-game performance scores. It does not provide deeper interpretation of learning patterns or automated insights that could support advanced instructional decision-making.
- The evaluation of the system is limited to internally generated pre-test and post-test assessments, in-game performance metrics, and user feedback collected during implementation. As such, results are dependent on system-generated data and are not validated against standardized external assessment instruments.

Review of Related Literature/Studies

The growing body of literature on game-based learning (GBL) and gamification in educational settings, particularly in history and social studies reflects a global shift toward interactive, technology-driven pedagogies. This review synthesizes related studies organized into five thematic clusters: (1) Problems and Challenges, (2) Key Features of GBL/Gamification Applications, (3) Research Methodologies Employed, (4) Developmental Tools and Frameworks, and (5) Recommendations from Existing Studies. The reviewed works span contexts in the Philippines, Southeast Asia, Europe, and internationally, covering K–12 and higher education levels.

Disengagement and Motivational Decline in History and Araling Panlipunan Education

A fundamental problem driving the adoption of GBL is the persistent disengagement of students in history and Araling Panlipunan (Social Studies) classes. Panga et al. (2025) reported that prior to gamification intervention, Grade 10 students at San Rafael National High School recorded a Mean Percentage Score (MPS) of only 44.44% in engagement-related indicators, classified as "Did Not Meet Expectations." Similarly, Florese (2025) observed that Araling Panlipunan is frequently perceived as content-heavy and challenging, driving low motivation among Grade 8 students in Paranaque. Despabiladeras et al. (2024) further noted that teaching Batas Militar (Martial Law) to elementary students in the 21st-century learning environment has proven to be a daunting pedagogical task due to the subject's sensitive nature and the lack of engaging instructional materials.

The kasaysayan study (Centeno et al., 2023) directly identified demotivation as a core problem, noting that students either find Philippine history difficult to understand or forget what they have learned over time. Pre-gameplay assessments showed a mean score of only 3.65 out of 9 points, with the majority of respondents scoring just 3 out of 9. Buenaflor (2024) similarly found that students' learning interests in the AGILA study were "not enduring and impactful" due to the limited range of modalities available, with most measured statements yielding only average results when using the modular approach.

A significant theoretical concern across the literature involves the overemphasis on extrinsic rewards. Scholars argue that point systems, badges, and leaderboards may reduce students' intrinsic desire to learn (Frontiers in Education, 2024). Ryan and Deci's Self-Determination Theory, cited across multiple studies including Panga et al. (2025) and meta-analyses by Zeng et al. (2024) and the Frontiers in Psychology meta-analysis (2023), posits that external rewards can undermine intrinsic motivation when they become the primary driver of engagement. Additionally, once novelty fades, students often lose interest if game mechanics are not dynamically updated or varied (Lai & Hu, 2025).

The adoption of GBL in developing countries, including the Philippines, is constrained by inadequate technological infrastructure, inconsistent internet access, and disparities in device availability. Buenaflor (2024) acknowledged that the limited range of modalities available contributed to the lack of sustained engagement. Barriers to gamification in developing countries (Sambo et al., 2025) include technological infrastructure gaps, equity concerns for students with disabilities or neurodivergent needs, and data privacy challenges arising from digital platforms collecting behavioral data without adequate safeguards. These systemic barriers risk widening existing educational inequalities rather than addressing them.

The Gaming Revolution study (Lai & Hu, 2025) highlighted that teachers often lack sufficient training in integrating GBL into formal curricula, and many face institutional resistance, time constraints, and limited access to professional development resources. A study on gamification in education confirmed that instructors in many institutions are only moderately prepared to use the gamification approach due to a lack of experience and time in developing gamification-based applications (Frontiers in Psychology, 2023).

A systematic review from early 2024 found that while gamified interventions generally positively influence academic performance and engagement, challenges in measurement and consistent implementation persist, making gamification only moderately effective overall. The meta-analysis by Zeng et al. (2024) covering 22 experimental studies from 2008 to 2023 found a moderately positive effect but noted significant variability in outcomes depending on game design elements, educational stage, and implementation fidelity.

Game Mechanics and Interactive Features That Enhance Historical Learning

Narrative-driven gameplay is among the most prominent features of Philippine history-focused educational games. Bayani Chronicles (Fernandez et al., 2025) employed quest-based storytelling and open-world exploration to transform Philippine history into immersive experiences. The game's narrative structure was organized around the lives and legacies of national heroes, providing historical context through interactive gameplay rather than passive instruction. Similarly, the Misterios de Olivenza serious game (Ramos et al., 2026) integrated historical, geographical, and cultural content related to Olivenza, Spain, through exploratory gameplay and problem-based activities using RPG Maker MV, demonstrating that teacher-created serious games with narrative depth can yield high motivation and perceived learning.

Interactive quiz and trivia mechanics appear across multiple Philippine-developed GBL applications. KaSAYSAYan (Centeno et al., 2023) was built as an Android-based adventure trivia game covering Philippine history topics aligned with the elementary curriculum's learner modules. The game's content included short-answer questions on Philippine history and enabled dual use as both a student learning tool and a teacher assessment instrument.

Gamification elements such as point systems, leaderboards, and reward mechanisms are central to many of the reviewed interventions. Panga et al. (2025) implemented weekly leaderboards and structured scoring for participation in Araling Panlipunan, which fostered healthy competition among Grade 10 learners. The AGILA study (Buenaflor, 2024) utilized game-based interactive learning activities where mean results based on students' capabilities in game-playing confirmed positive impacts on learning experiences. Despabiladeras et al. (2024) reported that incorporating game elements provided a more engaging, entertaining, and informative learning environment that promoted active participation and enhanced student achievement in learning Batas Militar.

More advanced GBL applications leverage three-dimensional environments to create immersive learning experiences. Bayani Chronicles (Fernandez et al., 2025) features a 3D

open-world setting where players explore environments tied to the stories of Philippine national heroes. Evaluation using the ISO/IEC 25010 quality framework showed particularly strong scores in usability, functional suitability, and educational value, achieving an overall "Above Average" rating. The National Historical Commission of the Philippines (NHCP) affirmed the game's historical accuracy and cultural relevance, making 3D design a key feature for ensuring both engagement and authenticity in Philippine history games.

Alignment with official curriculum content is a distinguishing feature of educationally effective history games. KaSAYSAYan (Centeno et al., 2023) directly referenced learners' modules already implemented in the elementary curriculum, ensuring content validity. The web-based learning application for Philippine history (ResearchGate, 2024) was similarly designed to enhance understanding of curriculum-specific historical content. Adaptive features that adjust to student performance levels and provide real-time data for teachers represent advanced design features. Research on digital educational games established that the digital learning environment moderates the relationship between game-based learning and motivation, and that engagement acts as a key mediating factor in how digital games drive student motivation (MDPI Education Sciences, 2023). These findings underscore the importance of designing games with feedback loops, performance dashboards, and adaptive difficulty.

Research Approaches in Evaluating Game-Based Learning Interventions

The most common research design in the reviewed quantitative studies is the quasi-experimental pre-test/post-test model. Florese (2025) employed a descriptive quasi-experimental design using a one-group pre-test and post-test model to assess the impact of digital game-based learning (DGBL) on 40 Grade 8 students at Paranaque National High School Baclaran, using a teacher-made achievement test and structured survey questionnaire. Despabiladeras et al. (2024) utilized a true-experimental research design, dividing 30 Grade 6 students into two instructional groups traditional and GBL and comparing pre-test and post-test scores using Paired t-Test and percentage frequency

distribution. Panga et al. (2025) employed a single-group research design with a five-week intervention, tracking engagement through structured scoring and weekly leaderboards.

Multiple reviewed studies employed meta-analysis and systematic review to synthesize broad evidence on gamification's effects. Zeng et al. (2024) conducted a comprehensive meta-analysis covering 22 experimental studies from 2008 to 2023 using a random effects model, revealing a moderately positive effect of gamification on academic performance (Hedges's $g = 0.782$). The Frontiers in Psychology meta-analysis (2023) examined the effectiveness of gamification as a tool for promoting teaching and learning across educational settings. A systematic review updated in early 2024 employed PRISMA 2020 guidelines and analyzed 394 studies from the Web of Science database across all education levels, identifying gamification's versatility across STEM, healthcare, and language learning contexts.

Studies that focus on the creation of educational game applications predominantly employ developmental research guided by Agile methodology. Bayani Chronicles (Fernandez et al., 2025) was guided by game-based learning theories and the Agile development model, employing developmental and experimental approaches with data collected from three respondent groups: students, teachers, and NHCP representatives. KaSAYSAYan (Centeno et al., 2023) similarly adopted the Agile methodology's five phases requirements, design, develop, test, and deployment as its development framework, using ISO 25010:2011 for software quality evaluation.

Many studies used structured questionnaires to measure students' perceptions, motivation, engagement, and self-efficacy. The serious game study by Ramos et al. (2026) employed a validated questionnaire to analyze gameplay experience, motivation, and self-perceived learning among 86 primary students, giving attention to age and gender differences and finding no statistically significant differences by gender. Buenaflor (2024) utilized an adopted and modified survey questionnaire on game-based learning attitudes to gauge participants' reactions and experiences after engaging in AGILA. The study involving Grade 9 students' perspectives on GBL in Social Studies employed qualitative data collection to explore student perceptions (ResearchGate, 2025). Panga et al. (2025) framed their study explicitly as action research, investigating gamification as an innovative

approach within the 2024–2025 school year, making it directly applicable to classroom practitioners seeking evidence-based strategies.

Tools, Platforms, and Frameworks in Developing Educational History Games

Unity is the most prominently used game development platform across Philippine-focused educational GBL studies. Bayani Chronicles (Fernandez et al., 2025) was developed using the Unity game engine, implemented via the Agile Software Development Life Cycle for iterative improvements. KaSAYSAYan (Centeno et al., 2023) was built on the Android platform using Unity, which enabled cross-platform development and rich interactive features. Unity's versatility supporting 2D and 3D development, C# scripting, and mobile deployment on Android and iOS makes it a natural choice for educational game developers targeting the Philippine K–12 market. Unity has also been used in international heritage education applications, including a 3D map-based Philippine heritage exploration app that integrated AI-driven NPCs and a gamified achievement system.

RPG Maker MV was used as the development platform in the Misterios de Olivenza serious game (Ramos et al., 2026), a teacher-created game for Social Sciences and heritage education. The study highlighted RPG Maker MV as an accessible tool for educators without advanced programming skills, enabling teachers themselves to design curriculum-aligned serious games. The findings demonstrated that teacher-created games built on accessible platforms can still achieve high levels of motivation and perceived learning, suggesting that tool accessibility is a critical factor in expanding the pool of available educational games.

The dominance of Android as the deployment platform across reviewed Philippine studies reflects the mobile-first digital landscape of Filipino learners. KaSAYSAYan (Centeno et al., 2023) was designed and deployed on Android, aligning with the reality that smart devices are popular and prevalent in the Philippine educational context. Hardware components commonly used in these projects include Android smartphones for testing and gameplay, and personal computers or desktops for development.

Multiple GBL development studies utilized the ISO/IEC 25010 quality framework for software evaluation. Bayani Chronicles (Fernandez et al., 2025) was evaluated using ISO/IEC 25010 across eight quality criteria including usability, functional suitability, performance efficiency, compatibility, reliability, security, maintainability, and portability achieving an overall "Above Average" rating with strong scores in usability and educational value. KaSAYSAYan (Centeno et al., 2023) applied the earlier ISO 25010:2011 standard for software quality evaluation, providing a standardized, internationally recognized basis for assessing the quality of educational applications developed as capstone or research projects.

The Agile Software Development Life Cycle, adopted by both Bayani Chronicles (Fernandez et al., 2025) and KaSAYSAYan (Centeno et al., 2023), supports iterative game development through cycles of requirements, design, development, testing, and deployment, allowing developers to incorporate feedback from users and educators at each phase. Beyond standalone game applications, some reviewed studies situate GBL within Learning Management Systems (LMS) or web-based environments. Despabiladeras et al. (2024) explored game-based learning through Learning Management Systems in teaching Batas Militar to elementary students, demonstrating that GBL can be integrated within institutional LMS platforms to reach broader student populations. A web-based learning application for Philippine history (ResearchGate, 2024) was developed specifically to enhance students' understanding of Philippine history through a web-accessible format, expanding accessibility beyond mobile-only platforms.

Recommendations from Existing Studies

Multiple studies recommend that educational games be designed in close alignment with official curriculum content to maximize educational value. Bayani Chronicles (Fernandez et al., 2025) suggests that future game-based applications seek official historical validation as it did through NHCP review to ensure content accuracy and cultural authenticity. KaSAYSAYan (Centeno et al., 2023) similarly recommends that game content be directly referenced from official learners' modules.

A consistent recommendation concerns the need to balance extrinsic game mechanics with intrinsically motivating design elements. The Frontiers in Education (2024) systematic review on gamification and school engagement recommends that educators use adaptive difficulty to personalize challenge levels, integrate stories that make learning personally relevant, and combine points and badges with opportunities for self-reflection. The meta-analysis by Zeng et al. (2024) advises careful consideration in the selection and utilization of game design elements, noting that outcomes vary significantly depending on educational stage and the nature of the gamification employed. Researchers broadly recommend designing gamification systems that satisfy students' psychological needs for autonomy, competence, and social connection.

Studies consistently recommend investing in teacher capacity-building for GBL integration. Lai and Hu (2025) recommend that educational institutions provide structured professional development to prepare teachers for curriculum-aligned game integration, and that teachers receive training to avoid common pitfalls such as overemphasizing extrinsic rewards or inconsistent feedback delivery. Platforms like Classcraft and Kahoot offer professional development communities that educators should leverage. Ramos et al. (2026) recommend adequate scaffolding and careful curricular integration to maximize the educational effectiveness of teacher-created serious games, suggesting that teachers need both technical training and pedagogical guidance.

Many reviewed studies acknowledge small sample sizes and short intervention periods as key limitations and recommend longitudinal, larger-scale research. Florese (2025) sampled only 40 students over a limited intervention period and recommends expanding the participant pool and increasing the duration of GBL implementation to better capture knowledge retention and long-term attitudinal changes. Panga et al. (2025) recommend extending the gamification intervention beyond five weeks and including multiple sections to enhance the generalizability of findings. The meta-analysis by Zeng et al. (2024) identifies the need for more experimental studies and recommends that future research include diverse educational stages, geographic contexts, and game design element combinations.

Recommendations for inclusive game design are prominent in the broader gamification literature. Sambo et al. (2025) recommend that all gamification systems adopt inclusive design principles to avoid leaving learners behind, particularly students with disabilities, neurodivergent needs, or limited technology access. Researchers further recommend that developers follow data protection laws and be transparent about how students' behavioral data will be used, stored, and shared. Equity and inclusion considerations are essential so that gamification does not widen existing educational gaps related to geography or financial situation.

Several studies recommend the integration of augmented reality (AR), virtual reality (VR), and artificial intelligence (AI) to further enhance the immersive potential of history-based GBL applications. Heritage Quest, referenced in related Philippine game development literature (2025), incorporated AI-driven NPCs and a 3D virtual map of Metro Manila, recommending that future applications leverage AI to create more responsive and personalized learning environments. Zeng et al. (2024) recommend that future research investigate how different game mechanics interact with learners' intrinsic motivation and social dynamics including collaboration versus competition—to better inform adaptive GBL system design. The MDPI Education Sciences journal (2023, 2026) further highlights AR and VR mobile applications as areas of growing promise, recommending that developers explore these technologies as the adoption of wireless networks and mobile devices lowers the barrier to entry.

Review of Related Systems

The Review of Related Systems (RRS) is a critical component of any capstone research project, providing a systematic analysis of existing technologies, platforms, and game-based learning systems that share conceptual, functional, or pedagogical similarities with the proposed study. This section reviews existing systems, the categories relevant to the development of a two-dimensional (2D) educational history game intended for use in the Philippine educational context. The systems examined in this review were selected based on their use of 2D or pixel-art visual formats, their integration of historical content or game-based learning mechanics, and their documented application in educational settings. Each

system is analyzed according to its developer, type, purpose, key features, strengths, limitations, and relevance to the proposed capstone project. This review aims to establish a foundational understanding of the current state of the field and to identify the gaps that the proposed system seeks to address.



Figure 1.1 Gameplay Interface and Street Scene of Pompeii: The Mystery of Lucius.

Danger in Pompeii (DNA Studios / Institut Le Rosey, 2024) Danger in Pompeii is a 2D adventure and educational game developed by DNA Studios in collaboration with Institut Le Rosey, based on historical documents about the eruption of Mount Vesuvius and everyday Roman life. Dna-studios the game features a point-and-click, side-scrolling 2D format in a comic-book art style drawn from historical sources, and is available on Steam, iOS, and Android. Its primary purpose is to promote history in an engaging, interactive manner. The game's educational content has been archaeologically verified and integrated naturally into all gameplay elements, allowing players to relive Pompeii's last day through quests set in authentic historical locations such as the Forum, Amphitheatre, and Thermal Baths. Dangerapompei An educational version supports classroom use, allowing educators to assign specific quests aligned with history and geography curricula, while also reinforcing language skills and digital literacy. Danger pompei In terms of game-based learning, the system exemplifies situated cognition players learn by doing within a

historically reconstructed environment aligning with constructivist pedagogical principles. However, its scope is narrowly limited to ancient Roman history and offers no formal in-game assessment system, no teacher analytics dashboard, and no content addressing non-Western or Southeast Asian history. Its relevance to a 2D Philippine history capstone lies in demonstrating how period-accurate art and quest-driven narrative can carry educational content without interrupting immersion.



Figure 1.2 Login Interface and Main Menu of Heroes of History by InnoGames.

Heroes of History: Epic Empire (InnoGames, 2023–present) is a mobile city-building strategy game developed by InnoGames that tasks players with constructing a thriving historic city while collecting heroes from various historical eras and mythological backgrounds. InnoGames The game spans from the Stone Age to the future and incorporates real historical figures such as Hercules and Napoleon as collectible characters with unique abilities. Google Play While the game uses history as a thematic backdrop, its

primary design purpose is entertainment rather than formal education. The historical content is largely cosmetic. Figures are used as gameplay units rather than as subjects of historical inquiry which means the game only superficially supports game-based history learning. Players and reviewers have noted that progression quickly becomes dependent on in-app purchases, limiting accessibility for learners. App Store There is no curriculum alignment, no integrated assessment, and no narrative grounded in historical scholarship. Its relevance to a history education capstone is limited, but it demonstrates the value of hero-collection mechanics and era-based progression as engagement tools that could be adapted into a more pedagogically structured 2D game.

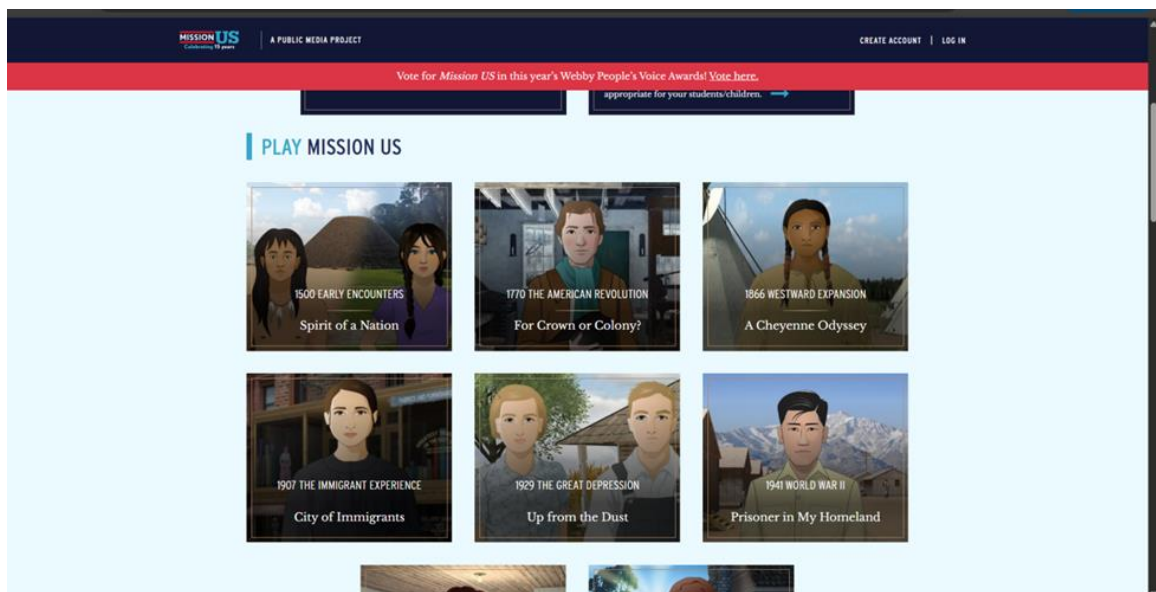


Figure 1.3: Episode Selection and Story Modules of Mission US by PBS.

Mission US (WNET/Electric Funstuff, 2010–present) is a deeply researched, award-winning educational media project developed for middle school students, designed to immerse players in historical and contemporary problems in ways that encourage perspective-taking, discussion, and weighing of multiple kinds of evidence. MissionUS The platform consists of eight online games in which players take on the roles of young people during critical moments in U.S. history, covering periods from colonial Boston in 1770 to the civil rights movement in Mississippi. Key features include immersive

role-playing, decision-making that shapes story outcomes, a teacher's dashboard for tracking progress and reviewing choices, and supplementary document-based activities and writing prompts. Ultimateradioshow The system strongly supports game-based learning theory, particularly through its branching narrative structure which develops historical empathy and critical thinking. Its effectiveness is well-documented through research, and it has garnered accolades including diverse perspectives often omitted in traditional history textbooks. Modulo Critically, however, Mission US is exclusively focused on American history and contains no content relevant to Philippine or Asian contexts. It also lacks a formal in-game scoring or assessment system, which limits its utility as a measurable learning tool in formal academic settings. Its teacher dashboard and curriculum-alignment model nonetheless provide a strong blueprint for integrating assessment and educator support into a 2D history game



Figure 1.4: Discovery Tour Time Periods Selection Interface of Assassin's Creed Discovery Tour.

Assassin's Creed / Discovery Tour (Ubisoft, 2007–present) series is widely regarded as one of the most popular franchises that bases itself on historical accounts, and has been praised as a case study in history classroom use due to its open narratives, situational immersion,

and ability to contextualize historical events through player avatars. ResearchGate Ubisoft further developed the Discovery Tour series dedicated educational, interactive, and violence-free experiences covering the Viking Age, Ancient Greece, and Ancient Egypt created in collaboration with historians and external specialists to ensure accuracy and suitability for educational use. Ubisoft Educationally, the franchise demonstrates that when learning objectives are blended seamlessly with gameplay rather than forced upon the player, learners are more likely to engage with historical content organically. Play the Past Despite these strengths, the series is a AAA, high-budget 3D open-world product with a significant hardware and cost barrier. It contains no formal in-game assessment, no localized content for Philippine history, and its complexity makes it unsuitable for younger or lower-resource learners.

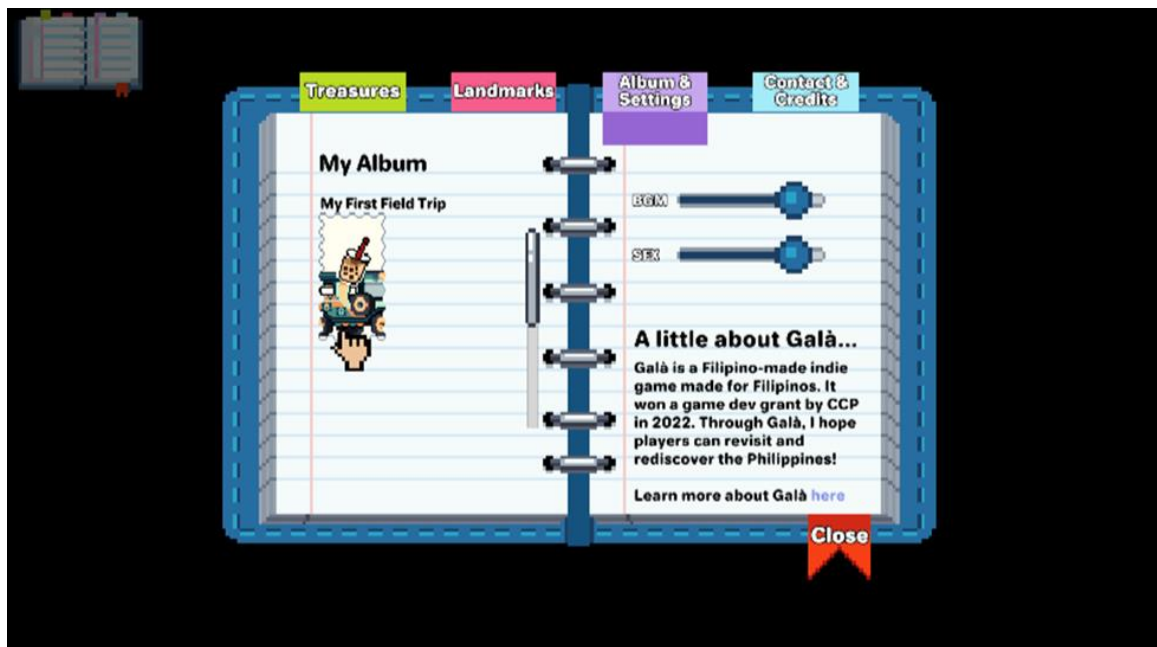


Figure 1.5: Album and Settings Interface of Galà, a Filipino Indie Educational Game.

Galà (Kendikorp, 2022) is a free-to-play mobile game by indie developer Kendikorp that takes players on a virtual road trip to explore various locations in the Philippines, learning about local history and culture along the way. Inquirer Players assume the role of a student traveling to different Philippine regions, collaborating with NPC classmates to discover

cultural artifacts and figures from Philippine mythology, such as the kubing jaw harp and the Bakunawa; the game is self-paced and does not use strict timers. Wikipedia Its developer, Audrey Stinson, stated that the goal is to "expose rather than educate," aiming to evoke a sense of wonder and curiosity about Filipino culture rather than delivering rote information. Newsbytes Funded by the Cultural Center of the Philippines' Game Development Grant, Galà is historically significant as among the first Filipino-developed games to center Philippine culture and history as its core subject. However, it lacks a structured quiz or assessment system, offers limited coverage of specific historical periods (e.g., colonial era, revolution), and has no teacher or classroom integration features. It is highly relevant to the proposed capstone as a precedent for locally developed, culturally grounded educational games, while also highlighting the gap in formal pedagogical scaffolding within the Philippine edu game space.



Figure 1.6: Player Information and Title Screen of Hiraya: A Bookworm's Odyssey.

Hiraya: A Bookworm's Odyssey (ViCaRy Inc.) A Bookworm's Odyssey is a Philippine history-based visual novel game situated in historical Philippine eras, wherein a player's

role is to acquire pages of a History Codex by answering quizzes in order to return to their present timeline. Itch The game was specifically developed for Grade 5 students as part of an academic research study, making it one of the few locally produced games directly targeting formal history education in the Philippine curriculum. Its 2D visual novel format uses text-based adventure and player choices as its primary mechanics. As a game-based learning tool, Hiraya directly embodies the use of narrative branching and embedded assessment (quizzes) to reinforce historical knowledge with a meaningful alignment with constructivist and inquiry-based learning frameworks. Its limitations, however, are considerable: the game's visual and interactive depth is minimal, it is confined to a narrow grade-level audience, it lacks multimedia richness (e.g., sound design, animated sequences, open exploration), and its distribution remains limited to itch.io as a research prototype rather than a commercially accessible product.



Figure 1.7: Character Profile and Historical Inspiration of Joe and Andrew in the Game.

Bayani: Fighting Game by Ranida Games is a 2D one-on-one fighting game inspired by Philippine historical figures, set in an alternate timeline with references to real historical

events. BAYANI Fighting Game It includes a relic collection mechanic that delivers bits of Philippine historical information as players progress. BAYANI: Fighting Game While it succeeds in promoting cultural awareness, it lacks curriculum alignment, formal assessment, and structured historical content, functioning more as entertainment than an educational tool.

Bayani Chronicles by Universidad de Manila is a 3D mobile game that presents the lives of Philippine national heroes through quest-based storytelling, open-world exploration, and mini-games, achieving an "Above Average" ISO/IEC 25010 rating with the National Historical Commission affirming its historical accuracy. Despite this, its 3D format caused performance issues on low-end devices Academia.edu, and its scope remains limited to national heroes with minimal narrative interactivity.

Both systems demonstrate strong cultural intent but share common gaps in accessibility, integrated assessment, and broader historical coverage gaps the proposed 2D history educational game capstone addresses through a lightweight format, embedded quizzes, and wider curriculum scope.

SYNTHESIS

The reviewed literature reveals that game-based learning (GBL) and gamification have become significant approaches in addressing the persistent issue of low student engagement in history and Araling Panlipunan education. Several studies consistently report that students perceive history as difficult, content-heavy, and less engaging, leading to poor academic performance and low motivation (Panga et al., 2025; Florese, 2025; Centeno et al., 2023). This disengagement underscores the need for more interactive and learner-centered teaching strategies, particularly in the Philippine educational context where standardized curricula and limited instructional resources continue to constrain teachers' ability to diversify their methodologies.

To respond to this issue, researchers emphasize the integration of interactive game mechanics such as narrative-driven gameplay, embedded quizzes, rewards systems, and

immersive environments which promote active learning and improve student motivation (Fernandez et al., 2025; Ramos et al., 2026; Despabiladeras et al., 2024). However, studies caution that excessive reliance on extrinsic rewards like points and badges may undermine intrinsic motivation if not properly balanced with meaningful and personalized learning experiences (Frontiers in Education, 2024; Zeng et al., 2024). Effective GBL design should therefore incorporate narrative relevance, adaptive difficulty, and opportunities for self-reflection to sustain long-term student engagement.

In terms of methodology, most studies utilized experimental and quasi-experimental designs to evaluate the effectiveness of GBL interventions, while others employed meta-analyses and systematic reviews to synthesize broader findings (Despabiladeras et al., 2024; Zeng et al., 2024). Overall, results indicate a moderately positive effect of gamification on academic performance and engagement, although outcomes vary considerably depending on game design quality, implementation fidelity, and the specific educational context in which the intervention was applied (Zeng et al., 2024). These findings highlight that GBL is not uniformly effective and requires deliberate instructional design to produce consistent results.

The literature also identifies commonly used development tools and frameworks, including Unity, RPG Maker MV, and Android platforms, alongside evaluation standards such as ISO/IEC 25010 (Fernandez et al., 2025; Centeno et al., 2023; Ramos et al., 2026). These tools support the development of interactive, accessible, and curriculum-aligned educational applications, particularly suited for the Philippine mobile-first learning environment where Android devices are widely prevalent among students and educators alike.

Despite these advancements, several limitations and challenges remain, including inadequate technological infrastructure, lack of teacher training, and issues related to accessibility and inclusivity (Buenaflor, 2024; Sambo et al., 2025; Lai & Hu, 2025). Moreover, many existing educational games lack integrated assessment systems, teacher analytics dashboards, and localized historical content gaps that are especially pronounced within the Philippine context. The reviewed related systems further confirm that while internationally developed games such as Mission US and Assassin's Creed Discovery Tour

offer strong pedagogical models, they remain inaccessible to Filipino learners due to their exclusive focus on Western historical content, high hardware requirements, and absence of curriculum alignment with Philippine standards.

Locally developed games such as *Galà* (Kendikorp, 2022) and *Hiraya: A Bookworm's Odyssey* (ViCaRy Inc.) represent meaningful strides toward culturally grounded educational gaming in the Philippines; however, both lack formal assessment systems and broader curriculum integration. Similarly, *Bayani Chronicles* (Fernandez et al., 2025) demonstrates strong educational intent and historical accuracy but is constrained by 3D performance issues on low-end devices and a limited scope focused primarily on national heroes. These gaps collectively justify the development of a lightweight 2D educational history game that combines interactive storytelling, embedded assessments, and localized Philippine historical content aligned with the official K–12 curriculum.

Overall, the reviewed literature supports the effectiveness of game-based learning in enhancing student engagement and learning outcomes in history education. It simultaneously underscores the urgent need for well-designed, curriculum-aligned, inclusive, and locally responsive educational games. The identified gaps in existing studies and systems, particularly the absence of integrated assessment, teacher support tools, and Philippine-specific historical content in accessible 2D formats directly inform and justify the objectives and design approach of the proposed capstone project.

METHODOLOGY

Technical Background

Technologies to be Used

The development of the MACARIO system integrates several technical components across different areas of software and web-based game development. Each technology was carefully selected to address the functional, educational, and accessibility requirements of the system. The following sections describe the core technologies used in the development of the platform, covering the type of application, programming approach, data management, server architecture, visual design, and audio integration.

The MACARIO system is developed as a web-based application accessible through any modern internet browser on Android mobile devices. This approach eliminates the need for device-specific installation or software downloads, as the application is delivered entirely through the web. By adopting a web-based platform, the system ensures broad accessibility across Android smartphones and tablets, making it available to a wider range of users regardless of their specific device model. Learners can access the application by navigating its URL, allowing institutions to distribute the system easily without technical barriers. The online nature of the platform also supports centralized updates, meaning improvements or content additions can be deployed without requiring users to manually update any installed software.

The system is built using the standard set of web development technologies, which form the foundation of all modern browser-based applications. The structural layer of the application is defined using a markup language that organizes and presents content within the browser, while a styling language is used to control the visual appearance, layout, and responsiveness of the interface across different screen sizes. The interactive and logical behavior of the game is implemented using a widely adopted scripting language that runs natively in web browsers, enabling real-time user interaction, game logic execution, and dynamic content rendering without requiring additional plugins or runtime environments. The programming approach follows object-

oriented and modular principles, allowing game components such as player controls, dialogue systems, level progression, enemy behavior, and scoring mechanisms to be developed as independent, reusable units. This structure improves code maintainability and supports future expansion of the system.

For data management, the system employs a combination of client-side storage and a server-side relational database to handle all user-related information. On the client side, the browser's built-in storage capabilities are used to temporarily retain session data, user preferences, and application states, reducing unnecessary server requests and improving responsiveness during gameplay. On the server side, a structured relational database is used to persistently store all critical data, including user accounts, game progress, quiz results, unlocked levels, and achievement records. This database is managed using a relational database management system that organizes data into structured tables with defined relationships, ensuring data integrity and efficient retrieval. The use of a server-side database also opens the possibility of future features such as educator dashboards, class-wide progress monitoring, and leaderboard functionality.

The system is built on a client-server architecture in which the web application is hosted on a remote server and delivered to users through the internet. The backend server is responsible for handling communication between the frontend application and the database, processing user requests, managing authentication, and serving game assets and content. A server-side web framework is used to define and manage these backend operations through a structured set of application programming interfaces, enabling organized and efficient communication between the client and server. This architecture allows the system to support multiple simultaneous users, maintain centralized control over application data, and ensure consistent performance and content delivery across different devices and geographic locations. Hosting the application on a server also simplifies maintenance, as system updates, bug fixes, and new content can be deployed centrally without any action required from end users.

The visual design of the MACARIO system is built around a two-dimensional digital art style that reflects the historical and educational theme of the game. All game assets, including character sprites, background environments, animated sequences, and user interface elements, are created

using a cartoons illustration approach. This style is chosen for its ability to evoke a historical atmosphere while keeping asset file sizes small and optimized for web delivery. The artwork is developed through a frame-by-frame and layer-based production workflow, which allows for precise control over character movements, environmental transitions, and interactive animations. Each visual element is designed with performance in mind, ensuring smooth rendering within the browser environment without placing excessive demands on device hardware. The cohesive visual style across all game scenes contributes to an immersive and engaging educational experience.

To further enrich the learning experience, the system incorporates a range of audio elements, including background music, ambient soundscapes, and interactive sound effects. These audio components are integrated into the web application using browser-native audio capabilities, which support playback of compressed audio formats optimized for web delivery. Background music establishes the historical and emotional tone of the game, while ambient sounds enhance the sense of place and immersion within each game's environment. Sound effects provide immediate feedback to player actions, reinforcing engagement, and making the gameplay experience more interactive and responsive. All audio assets are processed and compressed to minimize file size while maintaining acceptable quality, ensuring efficient loading and uninterrupted playback during gameplay.

Calendar of Activities

The development of the system follows a structured sequence of activities aligned with a simplified Rapid Application Development (RAD) approach. The timeline is organized into the following phases:

Completed and Ongoing/Upcoming

Task	February				March					April			
	1	2	3	4	5	6	7	8	9	10	11	12	13

Planning Phase													
Topic Ideation													
Title Proposal													
Analysis Phase													
Identifying the Problem													
Requirements Gathering													
Determining the Scope & Limitations													

Task	May				June					July			
	14	15	16	17	18	19	20	21	22	23	24	25	26
Design Phase													
System Design													
Game Storyline Development													
Level Design													
UI/UX Design													
Storyboarding													
Development Phase													

Gameplay Programming													
Character & Environment Creation													
Integration of Mechanics (combat, stealth)													
Scoring System Development													
Assessment Module Development													
Teacher Dashboard Development													
Database Integration													
Authentication System													
Sound Effects & Music Integration													
Mobile Optimization													

TESTING PHASE													
Functional Testing													
Bug Fixing													
Performance Testing													
User Testing (students)													
Task	August				September					October			
	27	28	29	30	31	32	33	34	35	36	37	38	39
Teacher Evaluation													
DEPLOYMENT PHASE													
Android Build Deployment													
Installation on Devices													
Initial System Run													
Task	November				December					January			
	27	28	29	30	31	32	33	34	35	36	37	38	39
Review													
Final Bug Fixing													

Mobile Optimization													
Student Feedback Collection													
Statistical Analysis													
Documentation Revision													
Final System Improvement													
Final Deployment													
Final Printing & Submission													

(Table 1.0 Gantt Chart)

Resources:

The development of the system requires the following resources:

Hardware:

The project will utilize the personal computers and laptops of the researchers. These devices must meet the following specifications to ensure smooth development, testing, and deployment:

Development Computer Specifications:

- CPU: Intel Core i5 (8th Gen) / AMD Ryzen 5 or higher
- RAM: Minimum 8 GB (16 GB recommended for better performance)
- Storage: 50 GB free (SSD preferred)
- GPU: Any dedicated GPU (GTX 1060 / RX 580+)
- Operating System: Windows 10/11, Ubuntu 20.04+, or macOS

Testing Devices:

- Android smartphones with at least 8 GB RAM
- Browser: Latest version of modern browsers such as (Google Chrome, ect.)

Software:

The proponents will utilize several software applications to ensure the successful development, design, and deployment of the system. Each software serves a specific purpose contributing to the project's efficiency and quality.

- Unity Primary development platform for the game's logic, interactions, and environment; supports 2D and 3D functionalities for augmented reality and gamified features.
- Frameworks/Libraries (optional depending on project) React.js / Vue.js / Angular for frontend development
- Backend Technologies Node.js / PHP / Python (Django or Flask) for server-side logic and MySQL / PostgreSQL / MongoDB – for database management
- XAMPP / WAMP / Localhost Server Used to simulate a local server environment for testing dynamic web applications.

- Aseprite Dedicated pixel art and sprite animation tool used to create all in-game visual assets including character sprites, tilesets, backgrounds, and frame-by-frame animations for the game.
- Audacity Free, open-source audio recording and editing software used to record, clean, and export all in-game sound effects including dialogue cues, environmental sounds, and action audio
- Pixilart is a free online pixel art tool that is easy for beginners and has a strong community for sharing and feedback, but it has limited advanced features compared to professional software.
- GitHub Version control and collaboration tool for synchronizing, tracking, and managing code changes using commands like push, pull, and merge.
- Microsoft Visual Studio Main Integrated Development Environment (IDE) for writing, debugging, and testing scripts implemented in Unity.
- Figma Used for designing UI/UX prototypes, planning navigation flows, and visual layouts before development.
- ChatGPT Used for assisting in code generation, debugging, documentation writing, and providing technical explanations during development.
- Canva /Adobe Photoshop
Used for creating graphical assets such as icons, banners, and interface elements.

Requirements Analysis

The system is designed to address the identified instructional challenges in Araling Panlipunan by analyzing the current learning environment based on the following factors:

Who (People Involved)

The system involves three primary groups.

Grade 8 students serve as the primary users who will directly interact with the web-based RPG as part of their learning experience. They are expected to access the system through internet-connected browsers on Android mobile devices, both during and after classroom instruction. The platform serves as a learning tool for reviewing and reinforcing concepts related to the life and historical role of Macario Sakay, particularly his leadership of the Tagalog Republic and sustained resistance during the Philippine-American War. Given that most Grade 8 students are already familiar with playing digital games and using online platforms in their daily lives, the system is designed to align with their digital habits and learning preferences through an engaging narrative-driven RPG format.

The Araling Panlipunan teacher plays the role of facilitator, guide, and evaluator. The teacher is responsible for integrating the web-based RPG into classroom instruction, directing students on when and how to use the platform, and assessing its overall effectiveness in supporting learning outcomes. The teacher may also monitor student progress and performance through a dashboard that displays act completion status, pre-test and post-test results, and in-game performance metrics such as objective completion, stealth effectiveness, and combat efficiency.

The system developers serve as designers and implementers, responsible for the full software development lifecycle from requirement gathering, system design, and coding, to testing, deployment, and maintenance. They ensure that the web application is pedagogically sound, technically functional, securely hosted online, responsive across Android devices, and aligned with the curriculum standards of the K–12 Araling Panlipunan subject, specifically the discussion of post-Philippine-American War resistance and Macario Sakay's role.

What (Activity)

The current instructional process in Araling Panlipunan primarily relies on lecture-based teaching, textbook reading, and memorization. These traditional approaches, while foundational, often result in limited student engagement, passive learning, and inconsistent retention of historical information, especially in complex topics such as the life of Macario Sakay, which involves a continuous narrative from Katipunan membership to guerrilla leadership and eventual capture. Visual aids such as charts and pictures are occasionally used, but they are not always sufficient to sustain student interest or promote deeper understanding of Sakay's historical contributions.

To address these gaps, the proposed web-based RPG aims to supplement existing teaching methods by providing a more interactive, narrative-driven, and engaging learning experience centered specifically on Macario Sakay. Through gameplay that includes stealth-based navigation, melee combat, dialogue-driven story progression, and mission-based objectives

across four narrative acts, the system seeks to make learning more dynamic and student-centered. Key activities include evading enemy patrols as a young Katipunan member, gathering intelligence for the Tagalog Republic, leading guerrilla raids, and experiencing Sakay's final surrender and trial. Before each act, a historical trivia fact activates prior knowledge, while pre-test and post-test assessments measure learning gains. This approach encourages active participation rather than passive reception of information, promoting both knowledge retention and critical thinking.

Where (Environment)

The system is designed to be used in two environments: inside the classroom during or after instruction, and at home for independent review and self-paced study. The primary platform is a fully online web-based RPG optimized for Android mobile devices, accessible through modern web browsers such as Google Chrome on Android smartphones and tablets. The system requires a stable internet connection for authentication, data synchronization, and accessing server-hosted game assets. Users must log in through the web platform to access their accounts, record their progress across the four acts, and synchronize their pre-test, post-test, and in-game performance data in real time. This ensures accurate monitoring of student engagement and learning outcomes across all Android devices.

When (Timing)

The system is intended to be used as a supplementary learning tool, not a replacement for classroom instruction. It is most applicable during lessons specifically covering the Philippine-American War and post-war resistance, allowing students to engage more deeply with Macario Sakay's story through interactive RPG gameplay. Since the platform is fully online, it can be accessed anytime and anywhere as long as there is an internet connection. Students can use it during class sessions, free periods, after school, or at home for review and practice. The four-act structure supports self-paced and differentiated learning, accommodating students who may need additional time to understand Sakay's historical narrative.

How (Current Procedure)

Under the current procedure, students receive historical information primarily through teacher-led discussions and written materials from textbooks or handouts. Assessment is often limited to written tests and recitations, which may not fully reflect a student's true level of understanding or engagement with the subject matter. These methods, while structured, do not always cater to varied learning styles or maintain sustained motivation among students. Specifically, the teacher notes that Macario Sakay's role is often mentioned only briefly due to time constraints, leaving students unaware of who he was and why he matters historically. The preliminary survey confirms this gap, with 81.4% of students unable to identify Sakay prior to the activity.

Proposed Improvement

The proposed system addresses these limitations by introducing an interactive, narrative-driven RPG that complements traditional instruction. By presenting Sakay's story through a game-based format on a familiar mobile web platform, the system aims to deepen historical comprehension in a way that lectures and textbooks cannot. Instead of generic quizzes or timelines, the system immerses students in Sakay's world through stealth missions, combat encounters, dialogue choices, and a progressively challenging difficulty curve across four acts. A historical trivia fact precedes each pre-test to provide context and activate prior knowledge. The teacher dashboard provides real-time access to student progress, act completion, and assessment results, enabling formative

evaluation without the pressure of formal examinations. This approach directly addresses the documented knowledge gap by presenting Sakay's life in an experiential, engaging, and memorable format.

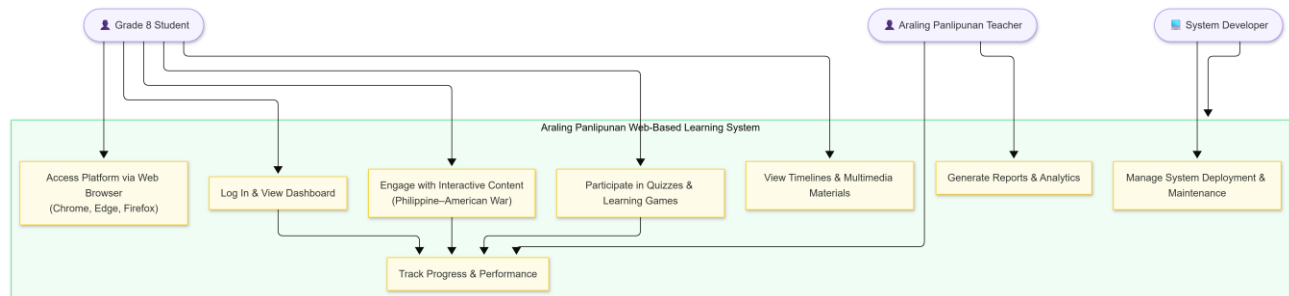


Figure 2.1 Summarize Using Use-Case Diagram

Requirements Documentation

The requirements for the MACARIO system were established through a preliminary data gathering activity, a consultation with the subject teacher, and an analysis of the current instructional challenges in Araling Panlipunan. The findings revealed a critical gap between traditional, lecture-based teaching methods and the learning preferences of Grade 8 students, who have demonstrated a strong preference for interactive and visual learning. The teacher, Ms. Kimberly Donadillo-Espiritu, highlighted difficulty in sustaining student interest in complex historical topics, specifically the life and role of Macario Sakay during the Philippine-American War. Additionally, 81.4% of surveyed students could not identify Sakay or describe his role, while 83.7% expressed greater interest in learning history through the perspective of a specific historical figure. Furthermore, 81.8% of students reported spending a significant portion of their free time playing digital games. This context established the need for a supplementary educational tool that transforms passive historical content into an interactive, narrative-driven RPG, delivered online to ensure seamless access and real-time progress tracking, without replacing the teacher's role.

Based on these discussions, the proposed system must support the following core requirements: a narrative-driven RPG structure that presents the life of Macario Sakay through four sequential acts (Act I: The Awakening, Act II: The Long Shadow of War, Act III: The Republic in the Shadows, Act IV: The Bitter Harvest); interactive player mechanics including stealth-based navigation, melee combat, dialogue-driven story progression, and traversal to reflect the contextual conditions of Sakay's guerrilla resistance; a dynamic difficulty system where mission complexity, stealth detection windows, and combat challenge increase progressively from Act I to Act IV; a health system that tracks player damage from enemy encounters and environmental hazards; an equipment system allowing players to equip items that modify combat effectiveness; a cosmetic reward system where earned in-game currency unlocks period-correct outfits; a trivia feature that presents a historical fact before each pre-test to activate prior knowledge; a teacher monitoring interface to track student progress through act completion, pre-test and post-test results, and in-game performance scores in real time; and full online functionality to enable immediate data synchronization and accessibility from any Android mobile device or PC with a web browser. The system is designed as a 2D side-scrolling web-based RPG optimized for Android mobile devices to ensure it remains lightweight and performant on student smartphones and tablets, directly addressing the students' familiarity with mobile gaming platforms.

The system must remain practical and understandable for its two primary user groups. For students, the interface must be simple, visually clear, and easy to navigate, allowing them to focus on Sakay's narrative and gameplay rather than complex controls. For the teacher, the monitoring interface must provide a straightforward view of class progress, act completion rates, and assessment averages without requiring advanced technical skills, as the current teaching environment already involves high instructional demands. The requirements were documented to establish a clear and mutual agreement between the developers and the end-users on what the software product must do to serve as an effective, engaging, and curriculum-aligned supplementary learning tool for Araling Panlipunan.

Functional Requirements

Requirement Name	Requirement Description
User Authentication	The system shall allow students and teachers to create accounts, log in, and access role-specific system functions.
Chapter Progression	The system shall present four sequential narrative acts (Act I: The Awakening, Act II: The Long Shadow of War, Act III: The Republic in the Shadows, Act IV: The Bitter Harvest) arranged chronologically and allow students to progress linearly through the game.
Player Movement	The system shall allow players to navigate the game environment using movement and jump controls.
Combat Mechanics	The system shall provide melee attack and special attack mechanics for combat-based gameplay interactions.
Stealth Mechanics	The system shall allow players to perform stealth-based actions to avoid enemy

	detection during designated missions.
Interaction System	The system shall allow players to interact with non-playable characters, dialogue prompts, and scripted objects within each act.
Narrative Delivery	The system shall present the life and historical role of Macario Sakay through dialogue, scripted sequences, and cutscene-based storytelling.
Dynamic Difficulty	The system shall progressively increase mission complexity, stealth detection windows, and combat challenge from Act I to Act IV.
Health System	The system shall track player health, depleting it upon taking damage from enemy attacks or environmental hazards, and allow health restoration through collectible items
Equipment System	The system shall allow players to equip items that

	alter combat effectiveness and survivability.
Cosmetic Reward	The system shall award in-game currency for completing objectives, allowing players to unlock period-correct cosmetic outfits.
Trivia	The system shall present a historical trivia fact related to Macario Sakay or the period before each act's pre-test assessment.
Act Assessment	The system shall administer pre-test and post-test assessments before and after each chapter.
Performance Scoring	The system shall compute in-game performance scores based on objective completion, stealth effectiveness, and combat efficiency.
Progress Tracking	The system shall record student act completion, assessment scores, and gameplay performance.
Teacher Monitoring	The system shall allow teachers to view student progress, act completion,

	assessment results, and in-game performance scores.
Data Synchronization	The system shall synchronize student progress and assessment results to the teacher monitoring interface upon internet availability.

Non-Functional Requirements

Requirement Name	Requirement Description
Performance	The system shall load gameplay acts, dialogue, and interface components within acceptable response time under normal operating conditions.
Reliability	The system shall consistently record and store student progress, assessment results, and gameplay performance during normal system use.
Usability	The system shall provide a user-friendly interface that is simple, clear, and understandable for Grade 8 students and teachers.
Accessibility	The system shall operate on Android mobile devices

	commonly accessible to student users.
Online Functionality	The system shall require an active internet connection for authentication, real-time data synchronization, and accessing server-hosted game assets.
Compatibility	The system shall be compatible with standard Android devices and screen sizes used in classroom settings.
Maintainability	The system shall follow a modular structure to support future updates, debugging, and content revisions.
Data Integrity	The system shall preserve student progress and assessment records without unintended data loss during normal operation.
Connectivity	The system shall require internet access for account authentication and synchronization of student results to the teacher monitoring interface.

acting and tailoring work as a disguise. The player also experiences a scripted social tension event where guards harass civilians, triggering Sakay's political awakening. The act concludes with a Katipunan initiation ceremony and an oath of entry. A historical trivia fact is displayed before the act's pre-test.

ACT II: THE LONG SHADOW WAR



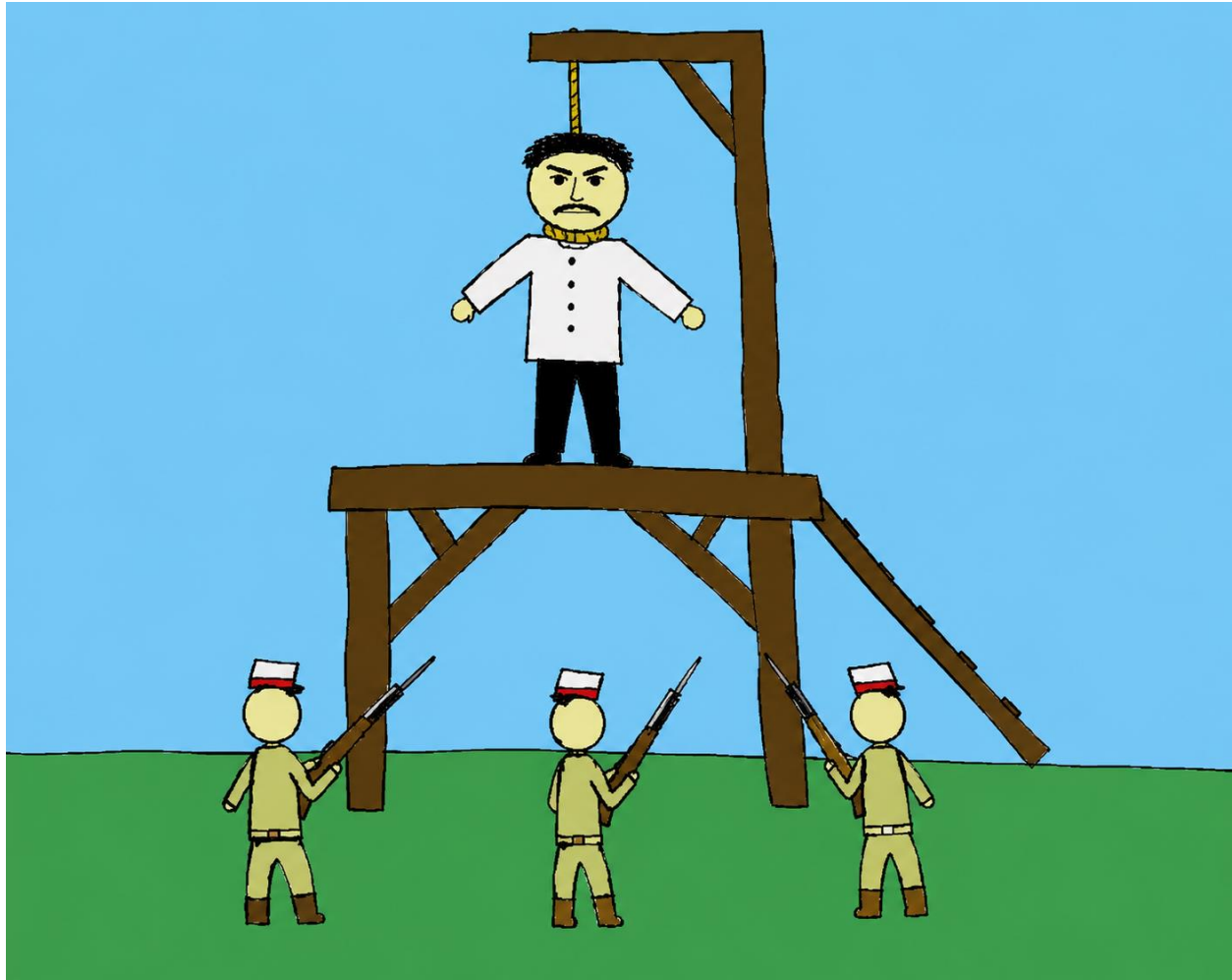
This act covers the outbreak of the Philippine-American War, the shift from Spanish to American rule, and Sakay's early resistance. Gameplay includes melee combat tutorials and mixed stealth and combat encounters with American patrols. The player navigates urban unrest during the revolt and experiences rising civilian tension as occupation replaces liberation. Following his capture, the player experiences a scripted release scene after the government declares an amnesty, allowing Sakay to return to the resistance. The act ends with Macario escaping and disappearing into hiding. A historical trivia fact is displayed before the act's pre-test.

ACT III: THE REPUBLIC IN THE SHADOWS



This act covers Sakay's retreat to the mountains, the establishment of the Tagalog Republic, and his guerrilla campaign. Gameplay includes open-terrain stealth navigation, enemy patrol avoidance, dialogue interactions with resistance leaders, and hit-and-run combat missions requiring tactical withdrawal after attacks. The player participates in a scripted declaration of the Tagalog Republic and assumes symbolic leadership roles. The act emphasizes guerrilla warfare tactics and survival in mountain environments. A historical trivia fact is displayed before the act's pre-test.

ACT IV: THE BITTER HARVEST



This act covers Sakay's surrender under false amnesty, his trial, execution, and legacy. Gameplay includes dialogue choices during negotiation with colonial intermediaries, a quiet approach to the meeting point, and a scripted betrayal leading to capture. The player then experiences a courtroom trial sequence where Sakay is condemned as a bandit, followed by his final speech (citing his actual last words, “Death comes to all of us sooner or later, so I will face the Lord Almighty calmly”) and execution through restrained visual framing. The act closes with a narrative recontextualizing Sakay as a revolutionary leader. A final historical trivia fact is displayed before the post-test.

Design of Software, System, Product, and/or Processes

The design methodology employs standard systems modeling techniques to describe the data flow, entities, and user interface of MACARIO. The following diagrams precisely capture the input-process-output logic, system context, internal data movement, local data structure, and key interface prototypes.

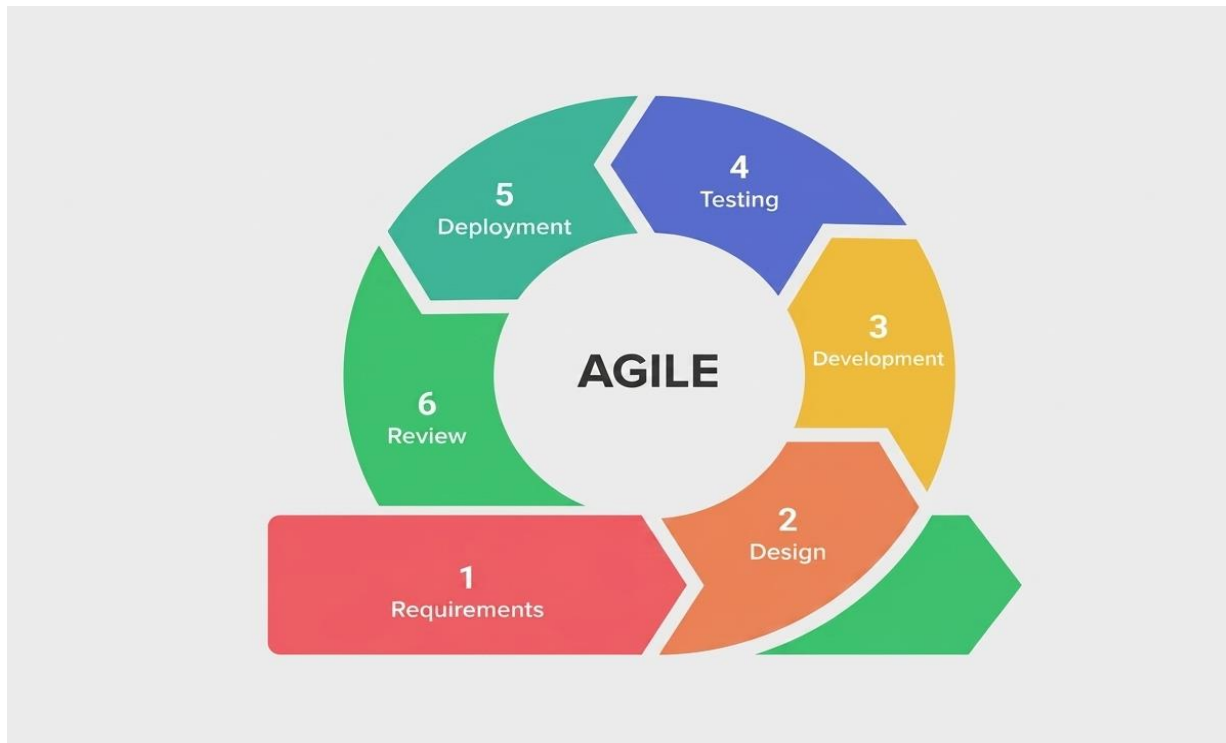


Figure 3.1 Agile Scrum Development

The development of MACARIO follows an iterative Agile methodology to support continuous improvement of both gameplay mechanics and educational content throughout the development process. The system is developed through repeated cycles of Requirements, Design, Development, Testing, Deployment, and Review, with each iteration producing incremental improvements based on evaluation and feedback.

In the Requirements stage, the researchers identify the core historical content to be included in the system, particularly the life and historical role of Macario Sakay during the Philippine-American War as aligned with the Grade 8 Araling Panlipunan curriculum. User needs are also determined, including student engagement requirements and teacher monitoring functions.

In the Design stage, the structure of the game is planned, including the linear progression of four narrative acts, gameplay mechanics such as movement, stealth, combat, health, equipment, dynamic difficulty, and detection systems, as well as the overall user interface for both students and teachers. Wireframes and system flow designs are also prepared during this phase.

In the Development stage, the system is implemented using the Unity game engine. Game mechanics such as player movement, enemy detection, health systems, equipment systems, and interaction systems are programmed, while narrative sequences, dialogue, and assessment components (including historical trivia before each pre-test) are integrated into each act as part of the ongoing build process.

In the Testing stage, the system is evaluated through functional testing of gameplay mechanics, assessment modules, and the teacher monitoring interface. Errors, inconsistencies, and gameplay balance issues are identified and recorded for revision.

In the Deployment stage, working builds of the system are made accessible through a web-based platform optimized for Android mobile devices for testing within controlled environments such as supervised classroom sessions or home use.

In the Review stage, feedback from testing is analyzed to identify areas for improvement, particularly in gameplay clarity, historical accuracy presentation, and system usability. Findings from this stage are used to refine and update the system for the next iteration.

This cycle is repeated until the system meets the research objectives and achieves acceptable functionality and usability for educational use.

Input-Process-Output (IPO) Diagram



Figure 3.2 Design of Software, System, Product, and/or Processes

The diagram illustrates an Input-Process-Output (IPO) model for a narrative-driven educational RPG about the life and historical role of Macario Sakay. An IPO model is a foundational systems engineering concept used to map out how a system functions by breaking down the data entering the system, how that data is transformed, and the final results produced.

The "Input" phase defines all the actions, commands, and existing data that feed into the game before any processing occurs. This includes real-time student interactions, such as selecting acts, moving characters, and engaging in combat or stealth. It also relies on background system data, including the player's profile, saved progress, and act objectives. Additionally, the input encompasses teacher commands via a connected platform, like logging in with a student ID or requesting to monitor class activity.

The "Process" phase is the core of the application where these inputs are analyzed, calculated, and transformed across three distinct engines. The game engine serves as the foundational software, translating movement and combat inputs into physical rules by executing core game logic and detecting collisions. Meanwhile, the narrative engine handles storytelling elements by

triggering cutscenes, advancing the plot, and displaying historical context pop-ups based on player actions. Simultaneously, a scoring engine works in the background to track completed objectives, compute performance scores, and store this progress data online.

The "Output" phase outlines the final results generated by the processing stage, delivering distinct feedback to both the student and the teacher. For the student, the output is the immediate player experience, comprising rendered game acts, narrative cutscenes, and audio feedback, alongside personal performance data like act scores and completed objectives shown on a results screen. For the educator, the system generates a separate teacher dashboard that aggregates the scoring engine's data to display student progress lists, act completion rates, and overall performance metrics to aid in class monitoring.

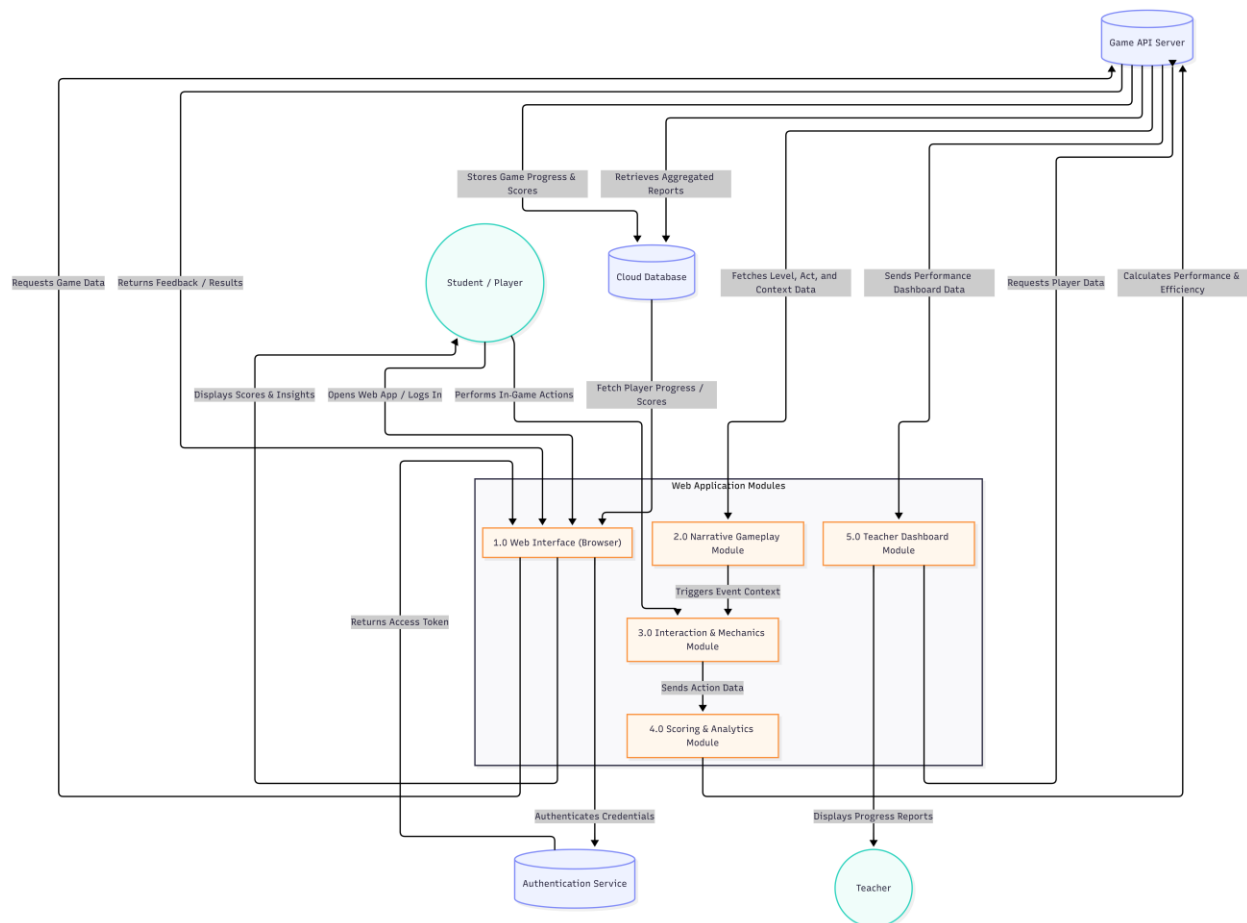


Figure 3.3 Data Flow Diagram

The DFD illustrates how data moves between the two external entities (the Student/Player and the Teacher) and the five internal modules of the MACARIO system.

The diagram presents the overall architecture and data flow of a web-based educational RPG designed to teach the life and historical role of Macario Sakay. At the center of the system is the Student/Player, who interacts with the application through a browser on an Android mobile device by logging in and performing game actions across four narrative acts. These actions are processed by different web application modules, including the Web Interface (Browser), Narrative Gameplay Module, Interaction & Mechanics Module, Scoring & Analytics Module, and the Teacher Dashboard Module. As the student plays, the system captures their actions, triggers game events (such as stealth, combat, dialogue, and act progression), and sends data to the scoring and analytics component, which evaluates performance based on objective completion, stealth effectiveness, and combat efficiency.

The system connects to a Cloud Database, where player progress across the four acts, pre-test and post-test results, unlocked cosmetic outfits, and other relevant data are stored and retrieved. This database also supports the generation of aggregated reports that can be accessed by teachers. The Game API Server acts as a central processor, handling requests such as fetching game data, calculating performance metrics, and sending processed information back to the application. Meanwhile, the Authentication Service ensures secure login by verifying user credentials and issuing access tokens.

On the teacher's side, the Teacher Dashboard Module retrieves processed data from the system and displays progress reports (e.g., act completion rates, assessment scores, and in-game performance insights). This allows teachers to monitor student learning outcomes effectively. Overall, the diagram shows how different components work together, from user interaction to data processing and reporting, to create an interactive, data-driven learning environment centered on Macario Sakay's historical narrative.

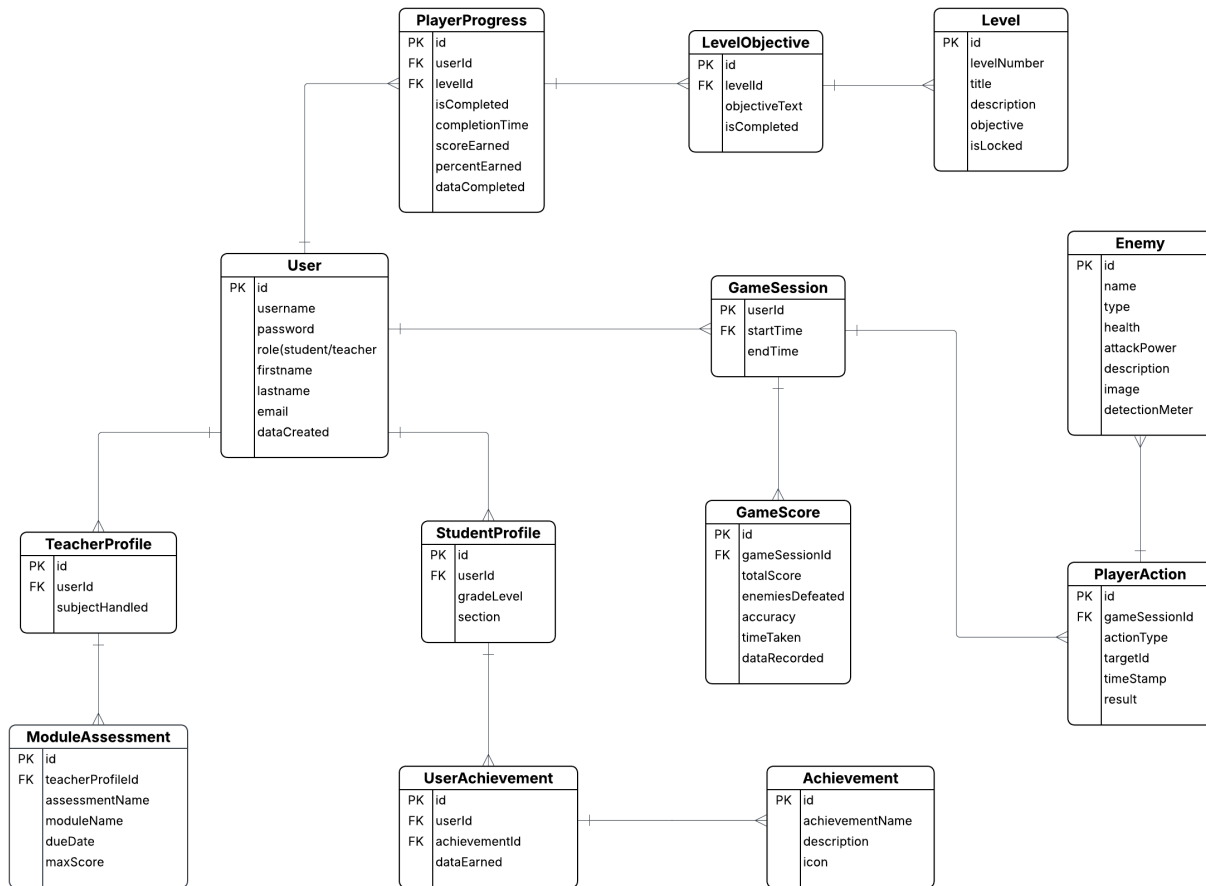


Figure 3.4 Entity Relationship Diagram

The ERD defines the complete data structure of the MACARIO system through fifteen interrelated entities (updated from thirteen to include Equipment and CosmeticReward).

The User entity serves as the central table, storing credentials and a role field that distinguishes students from teachers. It extends into two role-specific profiles: StudentProfile, which records the student's grade level and section, and TeacherProfile, which stores the subject handled. The TeacherProfile further connects to ModuleAssessment, which manages pre-test and post-test configurations per act, including due dates and maximum scores.

Gameplay activity is captured through the GameSession entity, which logs the start and end time of each session per user. Each session generates a corresponding GameScore record that stores the

total score, enemies defeated, stealth effectiveness, and time taken. Individual player actions within a session are recorded in `PlayerAction`, which tracks the action type (movement, combat, stealth, dialogue, equipment use), target, timestamp, and result, providing granular data on how each student engages with the game mechanics.

Act-related data is managed through the `Act` entity, which defines each of the four narrative acts (Act I: The Awakening, Act II: The Long Shadow of War, Act III: The Republic in the Shadows, Act IV: The Bitter Harvest), and the `ActObjective` entity, which stores the specific objectives tied to each act. Student progress across these acts is tracked in `PlayerProgress`, which records completion status, completion time, score earned, and percentage per act per user.

The `Enemy` entity stores the attributes of in-game antagonists, including health, attack power, and detection meter, which directly influence stealth and combat interactions during gameplay. The `Equipment` entity stores equippable items (e.g., melee weapons, defensive gear) that modify player combat effectiveness, while `PlayerEquipment` records which items each student has equipped. The `CosmeticReward` entity defines unlockable period-correct outfits, and `UserCosmetic` records which outfits each student has purchased using in-game currency.

Finally, the `Achievement` and `UserAchievement` entities handle the game's reward system. `Achievement` defines the available milestones (e.g., completing an act without detection, achieving perfect stealth, unlocking all outfits), while `UserAchievement` records which achievements each student has earned and when, supporting both engagement tracking and performance recognition.

Prototype (UI)

The following wireframes describe the core screens of MACARIO, ensuring alignment with the 2D pixel-art/cartoon, web-based design.



Figure 3.5 Prototype (UI) Gameplay



Figure 3.6 Login Screen



Figure 3.7 Assessment Screen



Figure 3.8 Progress Screen

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APPENDICES

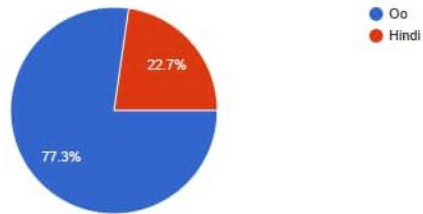
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Boring ba ang history subject (Kumpara sa ibang subject)?

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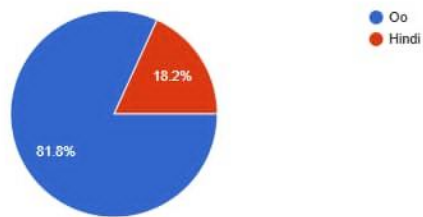
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Naglalaro ka ba ng video games on your free time?

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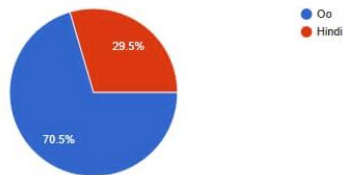
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Willing ka ba mag try ng interactive or visual learning methods?

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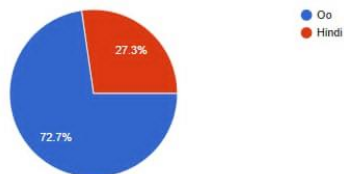
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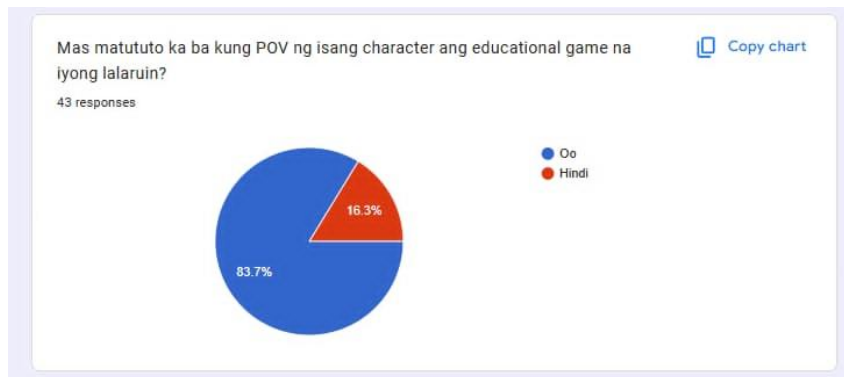
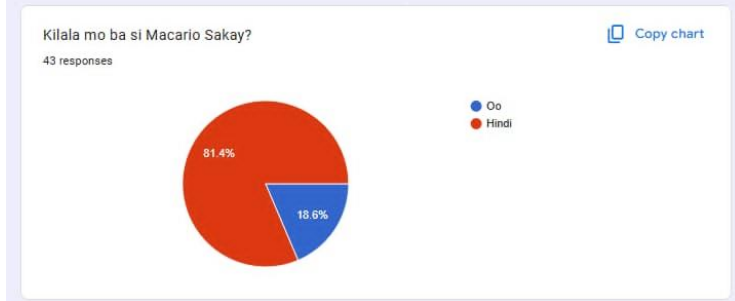
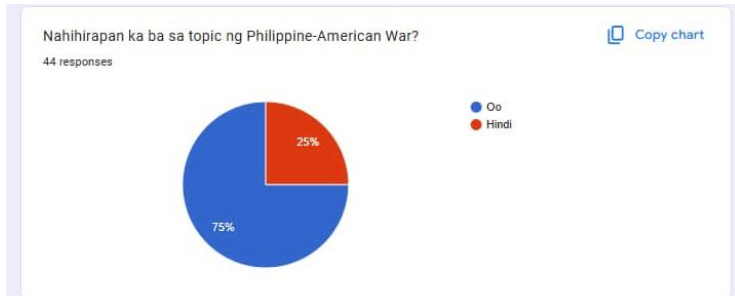


Nahirapan ka ba maalala yung mga lessons sa AP after ng klase?

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44 responses





APPENDIX A. RESOURCE PERSONS

INTERVIEWEE INFORMATION/CLIENT

Name: Kimberly Donadillo-Espiritu

Position: Araling Panlipunan Subject Teacher

Institution: Imus National High School

Role in the Study: Resource person and subject teacher consulted regarding the current challenges in teaching history and the effectiveness of existing instructional approaches in Araling Panlipunan.

INSTRUCTOR INFORMATION

Name: Felecisimo Buensuceso Jr., MIT

Position: Instructor

Institution: STI Dasmarinas

Role in the Study: Research instructor who guided the researchers in developing the study, improving the research structure, and ensuring the quality of the project proposal.

RESEARCH COORDINATOR

Name: Maureen Royce F. Sacay, MIT

Position: Research Coordinator

Institution: STI Dasmarinas

Role in the Study: Research coordinator who reviewed and monitored the progress of the study and provided academic recommendations for the completion of the project.

APPENDIX B. PERSONAL TECHNICAL VITAE

Curriculum Vitae of
Malulan, Kennel Keith L.

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0970-593-7723

EDUCATIONAL BACKGROUND

Level	Level of Competency	Name of School/Institution
Tertiary	2023 - Present	STI College of Dasmariñas
Vocational/Technical	2021 - 2023	Prenza National Highschool
High School	2017-2021	Prenza National Highschool
Elementary	2011 - 2017	Cabacungan Elementary School

SKILLS

Skills	Level of Competency	Date Acquired
JAVA	Beginner	2023
PYTHON	Beginner	2023
SQL	Beginner	2024
XML	Beginner	2025
CSS	Beginner	2026
HTML	Beginner	2026

Curriculum Vitae of
Riño, Kurt Vincent S.

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0919-308-8268

EDUCATIONAL BACKGROUND

Level	Level of Competency	Name of School/Institution
Tertiary	2023 - Present	STI College of Dasmariñas
Vocational/Technical	2021 - 2023	Asian Institute of Science and Technology
High School	2017-2021	Francisco E. Barzaga Integrated High School
Elementary	2011 - 2017	Dasmariñas Elementary School

SKILLS

Skills	Level of Competency	Date Acquired
JAVA	Beginner	2023
PYTHON	Beginner	2023
SQL	Beginner	2024
XML	Beginner	2025
CSS	Beginner	2026
HTML	Beginner	2026

Curriculum Vitae of

Castro, Chauncy John F.

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EDUCATIONAL BACKGROUND

Level	Level of Competency	Name of School/Institution
Tertiary	2023 - Present	STI College of Dasmariñas
Vocational/Technical	2021 - 2023	STI College of Dasmariñas
High School	2017-2021	Imus National High School
Elementary	2011 - 2017	Malagasang First Elementary School

SKILLS

Skills	Level of Competency	Date Acquired
JAVA	Novice	2023
PYTHON	Beginner	2023
SQL	Beginner	2024
C#	Novice	2025
CSS	Beginner	2026
HTML	Beginner	2026